Leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 2: exclude information unavailable in the intended prediction setting.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Choose a split unit that matches the intended prediction claim. Try the recall before opening notes.

Your learning target

I can choose a split unit that matches the intended **prediction** claim.

I can keep related examples together and audit **group** overlap.

I can record split roles counts and limits before model evaluation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

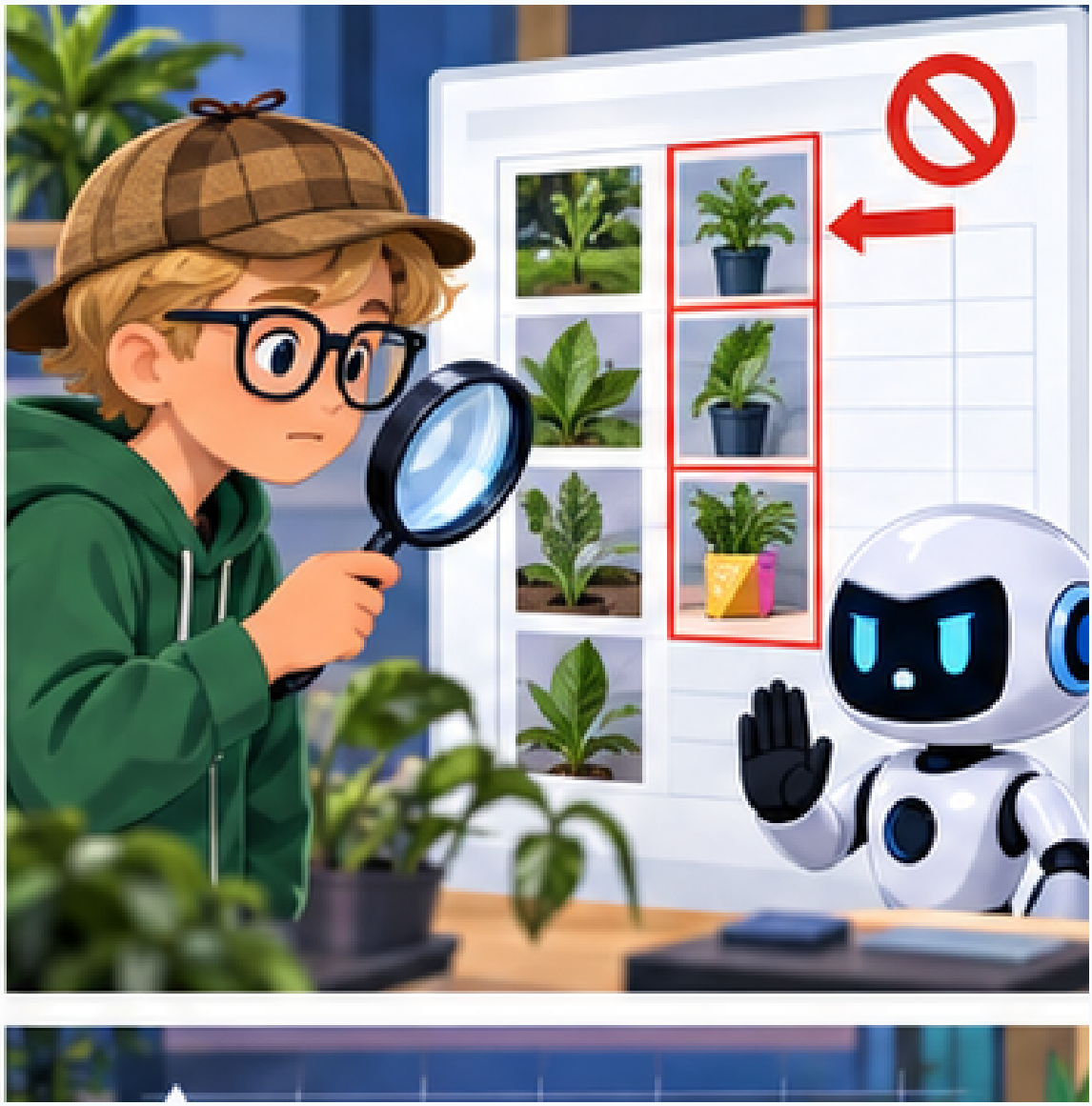
Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Leakage occurs when a feature exposes the target through information that would be unavailable, created later, or derived from the answer at real **prediction time**.



Your target

LEARNING TARGET Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

Builder name and experiment date:

Readiness for the connected explanation

Why does agreement between two annotators not prove their labels are correct?

Readiness for Scaling without leakage

What determines the nearest **training** neighbour, and which examples belong in the stored neighbour pool?

My readiness explanation

This week's end goal

By the end, I can explain **leakage** using a traced example and a stated limitation.

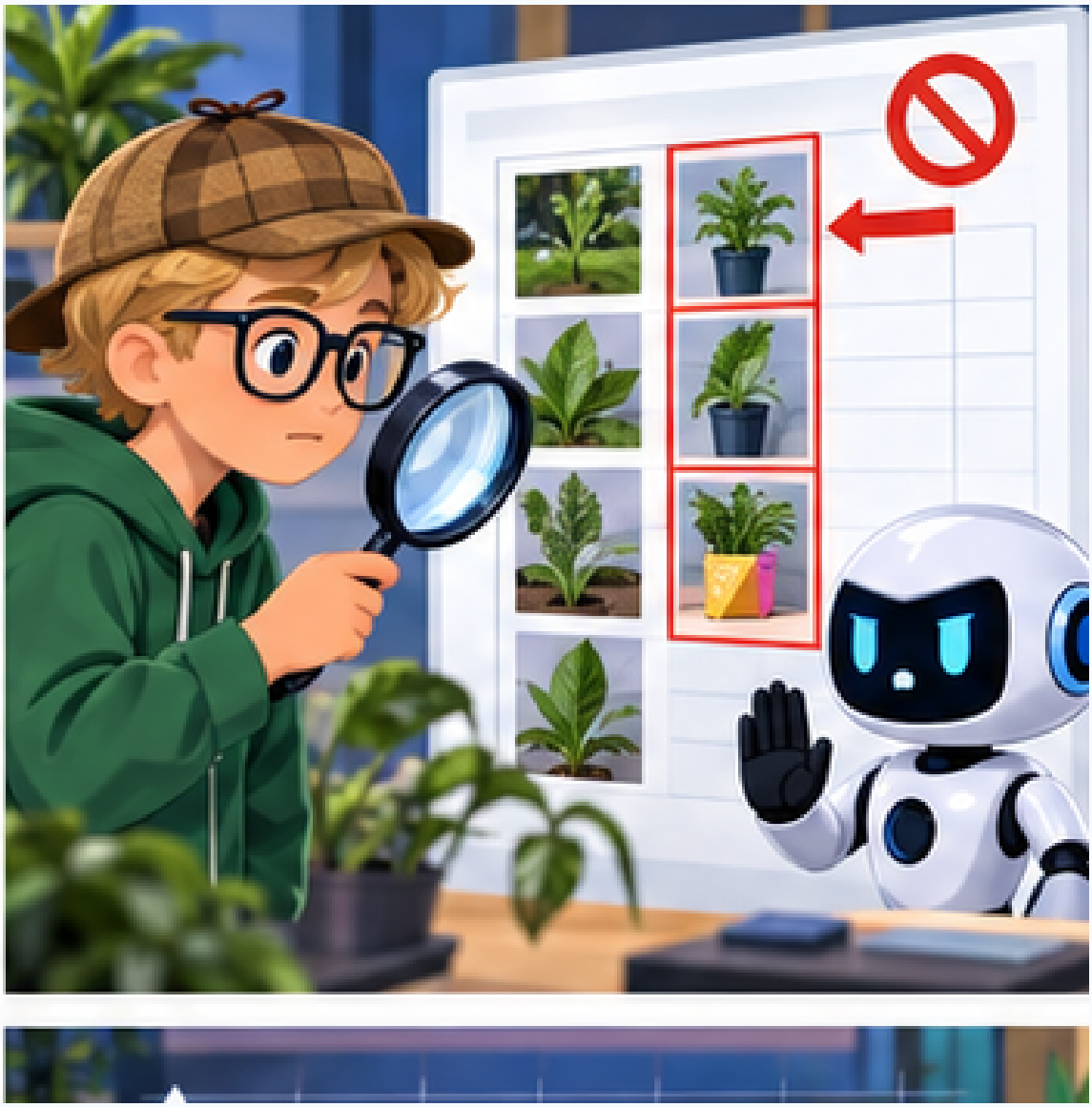
Notice the evidence

> Investigate the supplied Leakage case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **leakage** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A model may seem successful when its test examples closely resemble examples it already learned from. Imagine several pictures taken seconds apart during one robot trip.

They have different picture IDs but share a route, lighting and objects. This week you will assign entire trips to separate sets and explain what the resulting evaluation could measure. No model is fitted in this lesson.

Our invented twelve-row dataset contains frame descriptions from six trips. Each row is one frame; the **group** is its trip ID. The proposed claim concerns new trips on the same practice route under similar conditions.

We therefore keep all frames from a trip in one set. To claim success on new routes, **group**ing only by trip would not be enough.

Trip group	Frame IDs	Assigned role
A	A1 A2 A3	Training
B	B1 B2	Training
C	C1	Training
D	D1 D2	Validation

Trip group	Frame IDs	Assigned role
E	E1 E2	Test
F	F1 F2	Test

This small fictional assignment is for learning split design, not **evidence** of real performance. Each frame belongs to one **group** and one set.

Store frame and trip IDs as metadata for auditing; do not assume an ID is an appropriate **prediction input**.

Apply the connected idea: Scaling without leakage

Last week we compared one numeric feature. With two features, their numerical **range**s can give one feature much more influence. Scaling changes those contributions.

We will learn a minimum and **range** for each feature from **training** only, then reuse them for every example.

These fictional trips have a distance reading in cm, a brightness reading in sensor units and a reviewed lane label. Features are available before **prediction**; labels are separate **evidence**.

Related trips have already been **group**ed before splitting. For this exercise use k=1, add absolute differences across the two features, and break equal distances by alphabetical ID.

ID and label	Distance cm	Brightness units
A clear	10	200
B blocked	20	100
C blocked	20	200

For query (11,120), raw distances to A,B,C are 1+80=81, 9+20=29 and 9+80=89. B wins and predicts blocked. Adding these different units is an unscaled numerical choice; the large brightness **range** strongly affects it.

Scaling offers another explicit distance choice, not proof of which features matter.

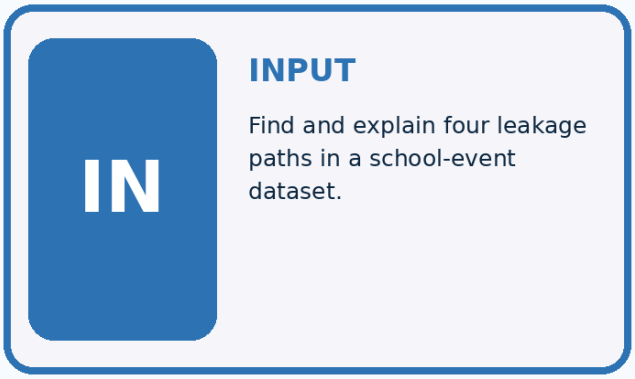
See how it works

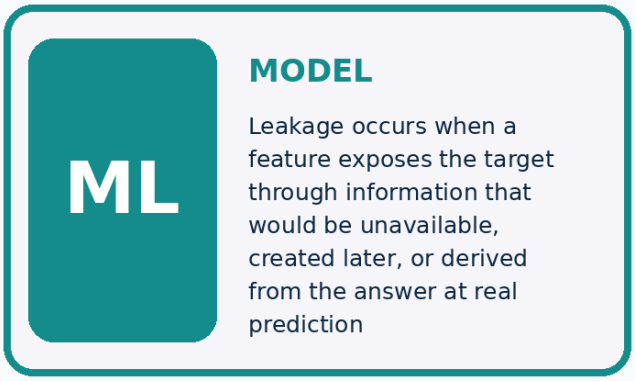
> Follow the mechanism for Leakage. Connect each labelled input, deciding step and result.

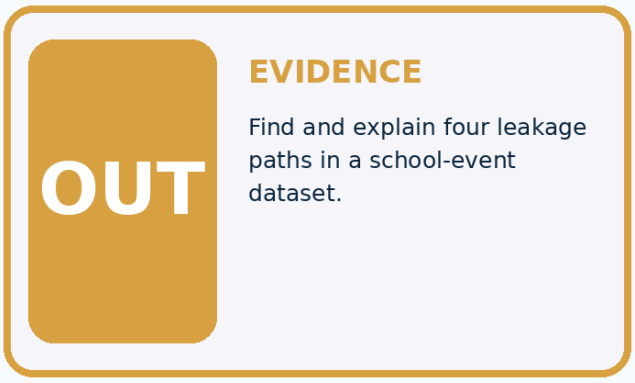
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Find and explain four **leakage** paths in a school-event dataset.

MODEL IDEA **Leakage** occurs when a feature exposes the target through information that would be unavailable, created later, or derived from the answer at real **prediction time**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD A leaf file name contains 'diseased'—the answer is hidden in the **input**. Treatment outcome was recorded after diagnosis, so it is unavailable at **prediction time**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Training data fits the model. Validation data helps compare choices such as candidate rules. Keep the **test set** for a final check after those choices are fixed.

A learner who repeatedly changes the model after seeing test results has used that set for development; its score is no longer an untouched final check.

```
train_groups = {"A", "B", "C"}
valid_groups = {"D"}
```

```
test_groups = {"E", "F"}
shared = ((train_groups & valid_groups)
          | (train_groups & test_groups)
          | (valid_groups & test_groups))
print(sorted(shared))
print(len(train_groups), len(valid_groups),
      len(test_groups))
```

The first line of **output** is []: no trip ID is shared between any pair of sets. The second is 3 1 2: **training**, validation and test contain three, one and two trip **group**s. The symbol & finds shared values and | combines sets.

These are **group** counts, not frame counts.

Keep each trip in one role

WHOLE GROUPS	ROLE	PURPOSE
A B C with all frames	Training	Fit from examples
D with all frames	Validation	Compare choices
E F with all frames	Test	Final check after choices

Keep related frames together. Separate group IDs prevent this form of overlap; they do not prove that every other source of leakage is absent.

This audit assumes the lists correctly describe every frame assignment. It cannot find an omitted frame, a wrong trip ID, a copied image with a different ID or a feature containing the answer.

Check the assignment table and source records too.

Apply the connected idea: Scaling without leakage

```
train = [(10, 200), (20, 100), (20, 200)]
low = [min(row[j] for row in train) for j in range(2)]
span = [max(row[j] for row in train) - low[j]
        for j in range(2)]
if 0 in span:
    raise ValueError("Review a constant training feature")

def transform(row):
    return tuple((row[j] - low[j]) / span[j]
                 for j in range(2))

scaled_train = [transform(row) for row in train]
query = (11, 120)
print(low, span)
print(scaled_train)
print(transform(query))
```

For each feature use scaled value = (value minus **training** minimum) divided by **training range**. **range**(2) visits feature positions 0 and 1.

The code prints [10, 100] [10, 100], then [(0.0, 1.0), (1.0, 0.0), (1.0, 1.0)], then (0.1, 0.2). Each minimum becomes 0 and each maximum becomes 1.

The first calculations fit low and span from train. **transform** uses those stored values without learning new ones. The code assumes nonempty **training** rows with exactly two finite numeric features; it does not check all **input**s.

Its explicit zero-**range** guard stops division by zero so a constant feature can be reviewed. This is a classroom policy, not a claim that every library stops this way.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

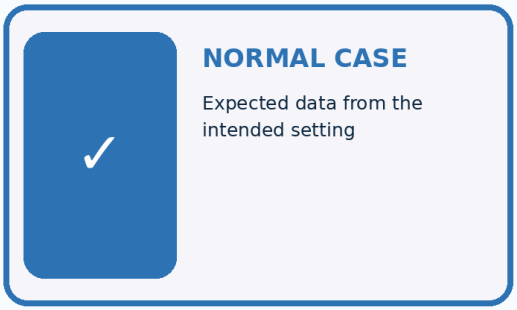
Compare the cases

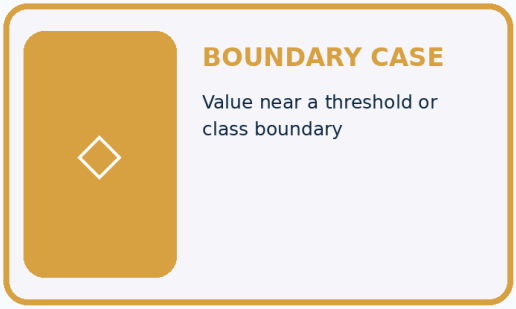
> Compare the supplied cases for Leakage. Identify the detail that changes the conclusion.

EVALUATION DESIGN

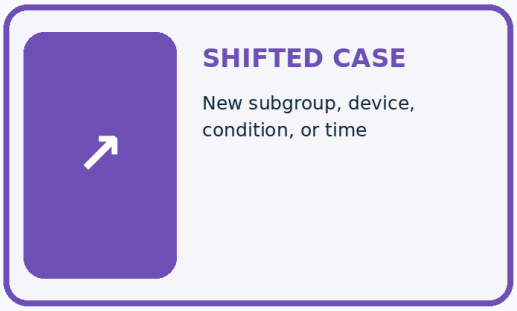
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, sub**group**, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Move Fir's first view into **training** while retaining its other two in test.

Predicted result

Prediction: The unseen-plant evaluation is compromised.

Why the result follows

Reason: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Leakage to explain a concrete decision, rather than only memorising definitions.

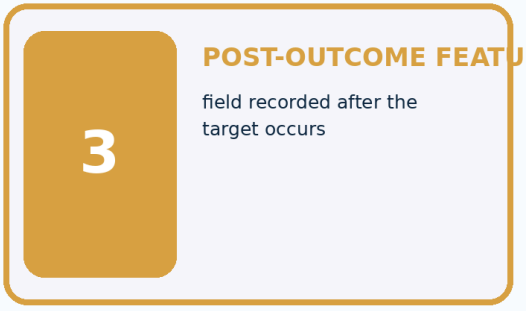
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
leakage	target information improperly enters features	
prediction time	moment when real input is available	
post-outcome feature	field recorded after the target occurs	
split leakage	related records cross data splits	

MISCONCEPTION TO REPAIR A highly accurate feature should be kept even when it would not exist during real use.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Training set	Examples used to fit a model
Validation set	Separate examples used to compare choices during development
Test set	Held out examples used for final evaluation after choices are fixed
Group	Related examples that must stay together for a stated evaluation goal

Terms for Scaling without leakage

Term	Meaning
Range	The largest training value minus the smallest for one feature
Fit	Learn and store values from training data
Transform	Apply a stored calculation to feature values
Leakage	Use information that should be unavailable while developing or evaluating a prediction method

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Leakage, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 8 • WORKED EVIDENCE

1. A leaf file name contains 'diseased'—the answer is hidden in the input.
2. Treatment outcome was recorded after diagnosis, so it is unavailable at prediction time.
3. Near-duplicate photos of one leaf appear in train and test.
4. Remove the leaking fields, group related photos before splitting, retrain, and reevaluate.

Step 1

STEP 1 A leaf file name contains 'diseased'—the answer is hidden in the **input**.

Step 2

STEP 2 Treatment outcome was recorded after diagnosis, so it is unavailable at **prediction time**.

Step 3

STEP 3 Near-duplicate photos of one leaf appear in train and test.

Step 4

STEP 4 Remove the leaking fields, **group** related photos before splitting, retrain, and reevaluate.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Worked explanation · Leakage

Suppose A1 and A2 are in **training** while A3 is in test. The frame IDs differ, but A appears in both sets. Shuffling the rows does not remove this relationship.

For our new-trip claim, move all A frames to one chosen role and record the revised assignment before evaluating a model. Do not copy A3 into both roles.

Worked explanation · Leakage

Audit that every source frame occurs exactly once, no **group** crosses roles and the assigned counts sum to the source count. In our worked plan the row counts are six **training**, two validation and four test: twelve total.

A fixed random seed can reproduce a random split, but it cannot make a flawed **group**ing choice fair.

A learner moves D2 from validation to training but leaves D1 in validation. Give the new row counts and shared trip ID. Explain why different frame IDs and a fixed seed do not repair the overlap.

Worked explanation · Leakage

A split record names the **prediction** goal, source version, **group** definition, exact assignments, row and **group** counts, date and any random method or seed used.

For this hand-assigned example, record that there was no random sampling. Keep label-rule versions and unresolved review statuses traceable as in Week 5.

Apply the connected idea: Scaling without leakage

One stored scale for every split

EXAMPLES	OPERATION	STORED VALUES
Training only	Fit minima and ranges	low 10 100 span 10 100
Training examples	Transform each row	Reuse low and span
Validation and new	Transform each row	Reuse low and span

Only training examples determine the scale. Transform validation and new examples using the same stored feature order, minima and ranges.

Worked explanation · Leakage

The scaled query is (0.1,0.2). Distances to scaled A,B,C are 0.1+0.8=0.9, 0.9+0.2=1.1 and 0.9+0.8=1.7. A now wins and predicts clear. We changed feature contributions; we did not change any reference label.

Compare scaled queries with scaled **training** rows, never with raw **training** rows.

Show why raw distance selects B but scaled distance selects A for (11,120). Which stored numbers must remain fixed when the next query arrives?

Equal **training range**s after scaling do not establish equal real-world importance. Outliers affect minima and **range**s. Scaling does not repair wrong labels, missing information or an unsuitable distance measure.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Leakage

Split by plant identity when testing generalisation to new plants.

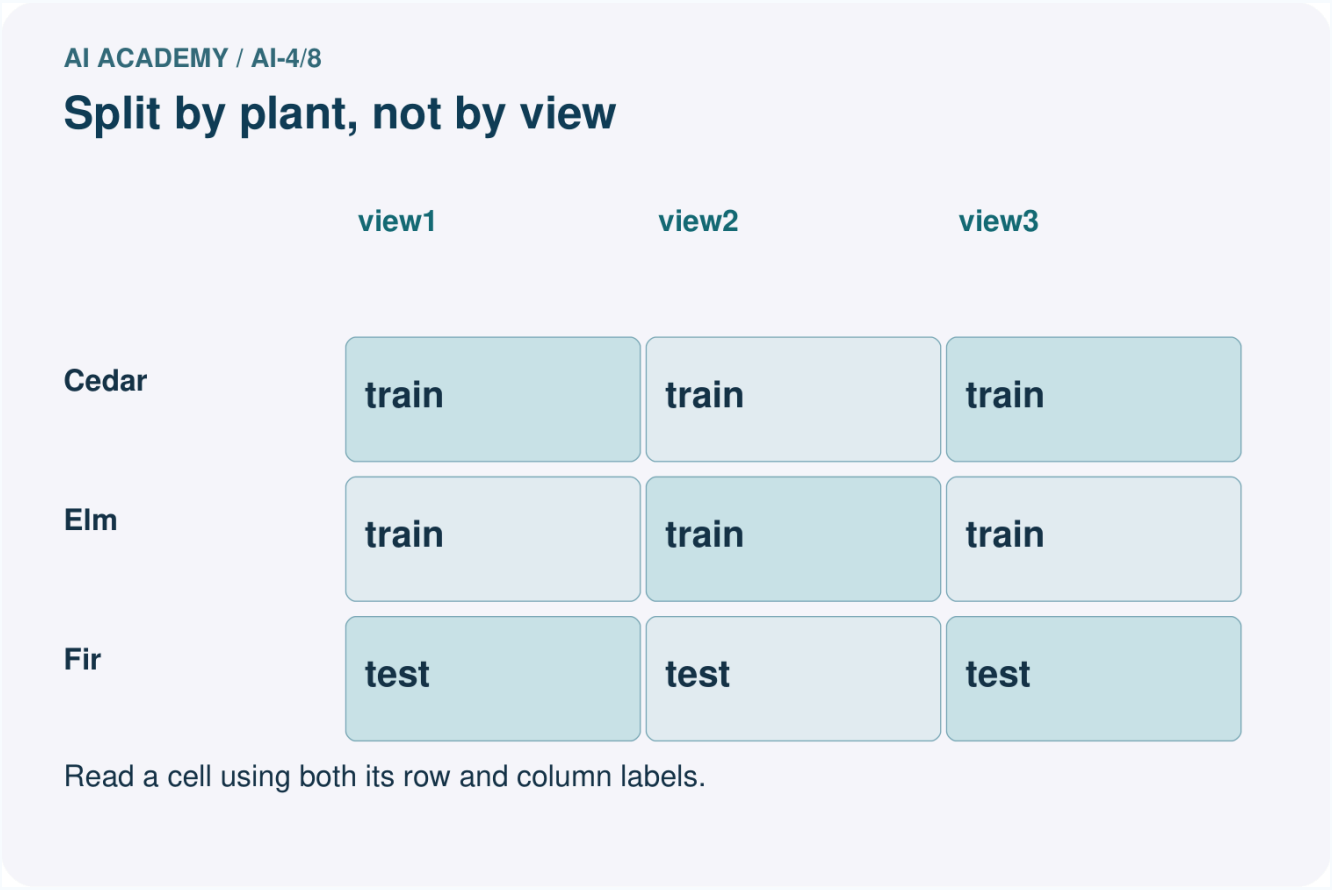
The facts we will use

Supplied facts

Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/8

Leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group related views

All three Cedar views stay together; likewise Elm and Fir.



2. Assign groups

Training receives six views from Cedar/Elm; test receives three Fir views.



3. State claim

The final test probes one unseen plant, not three independent plants.

CHANGE ONE THING

Move Fir's first view into training while retaining its other two in test.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group related views	All three Cedar views stay together; likewise Elm and Fir.
2. Assign groups	Training receives six views from Cedar/Elm; test receives three Fir views.
3. State claim	The final test probes one unseen plant, not three independent plants.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A correct **group** split still needs enough representative plants; this tiny exercise is not a

reliable population estimate.

Keep the stated limit

Read the labelled picture: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Move Fir's first view into **training** while retaining its other two in test. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Leakage claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A highly accurate feature should be kept even when it would not exist during real use. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A highly accurate feature should be kept even when it would not exist during real use.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

Fit learned pre**process**ing, such as scaling values using an average, on **training** data and apply the stored **transform**ation to the other sets.

Using the full dataset to learn such values would let held out information influence development. We will implement pre**process**ing later; **group** separation alone does not solve that issue.

Check the claim against the facts

Record exclusions and label-review policies before scoring. Do not remove difficult test cases because they reduce the score.

If a genuine data error requires repair, document it and its effect on the evaluation rather than claiming the old test stayed untouched. Next week you will build a simple baseline using these split roles.

All trip IDs are separate, but every trip uses the same route. Can you claim the model works on unseen routes? Explain the limitation and propose a split unit for that new claim.

Apply the connected idea: Scaling without leakage

Check the claim against the facts

Record the split, feature names and units, feature order, minima, **range**s, constant-feature policy, distance, k and tie rule. Keep the scale with the stored **training** examples. New data uses the same calculation.

If development choices change, document them and reserve a separate final test. Next week combines these steps into one reproducible classifier pipeline.

Check the claim against the facts

If a feature is constant in **training**, its **range** is zero. This example stops for review.

One documented option is to omit that feature everywhere and update the recorded feature list; do not divide by zero or invent a scale using validation values.

Library policies differ, so inspect and record the one actually used.

A learner finds minima and ranges using all rows, then splits them and says there is no leakage because labels were never used. Explain the error and give the correct order of operations.

STUDENT LESSON · 8 OF 18

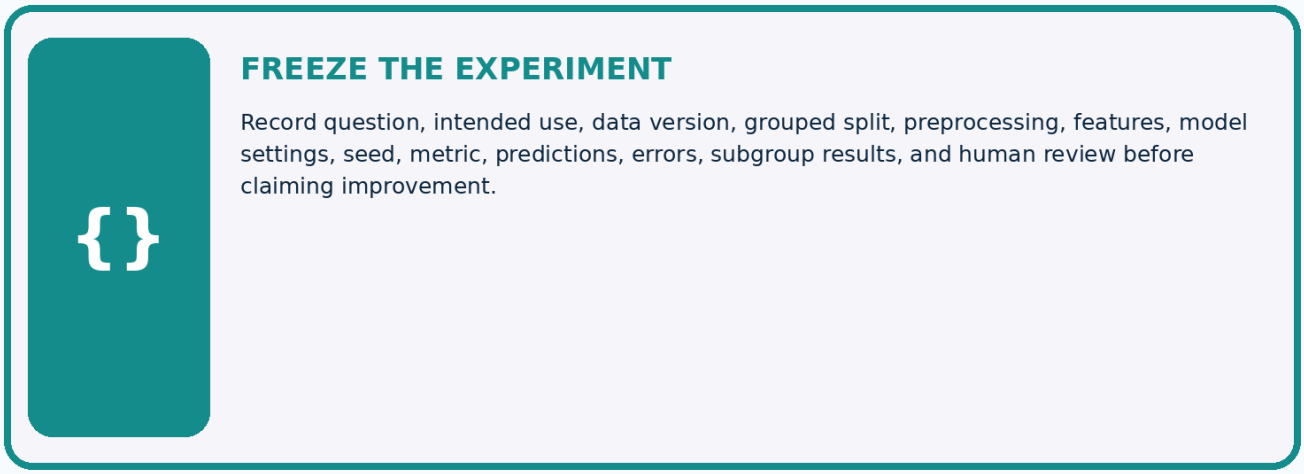
Practise together

Practise Leakage with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + **group**ed split + **leakage** check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + sub**group** slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

For new trips on a familiar route, holding out whole trips addresses within-trip similarity. For new routes, hold out entire routes so all their trips stay together.

For forecasting future events, a time-respecting split may be needed: **training** on later events to predict earlier ones would not represent that intended use. The evaluation question determines the separation.

There is no universal percentage that makes a split valid. Large **group**s can make row counts unequal.

Check that the sets can support the intended comparison, including enough relevant cases and labels, without moving individual related frames just to hit a percentage.

This twelve-row example is too small to justify a reliable performance claim.

Apply the connected idea: Scaling without leakage

Keep the **training** scale and k=1 fixed. Validation rows are (11,120,clear), (19,190,blocked), (12,80,clear). Their scaled feature pairs are (0.1,0.2), (0.9,0.9), (0.2,-0.2).

Nearest IDs are A,C,B, so **prediction**s clear,blocked,blocked score 2/3. The **training**-majority blocked baseline gets 1/3. These three examples do not establish general superiority.

The last brightness value scales to -0.2 because 80 is below the **training** minimum 100. New values can fall outside 0 to 1; do not refit or silently clip them.

Including validation features in minima or **range**s leaks evaluation information even without labels. **Fitting** a separate scale on validation also changes coordinates and breaks the intended comparison.

Practise with less help: Distinguish frames from groups

Use workbook W2 for the connected example “Splits and related examples”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Scaling without leakage

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before **group** discussion: Every **transform**ed value must lie between 0 and 1

Explain the picture

How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Leakage. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Leakage Detective: draw a timeline for every suspicious field, inspect group identity, and repair the split before training.

1. Write the hypothesis and expected **evidence** before running.

2. **Leakage** Detective: draw a timeline for every suspicious field, inspect **group** identity, and repair the split before **training**.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

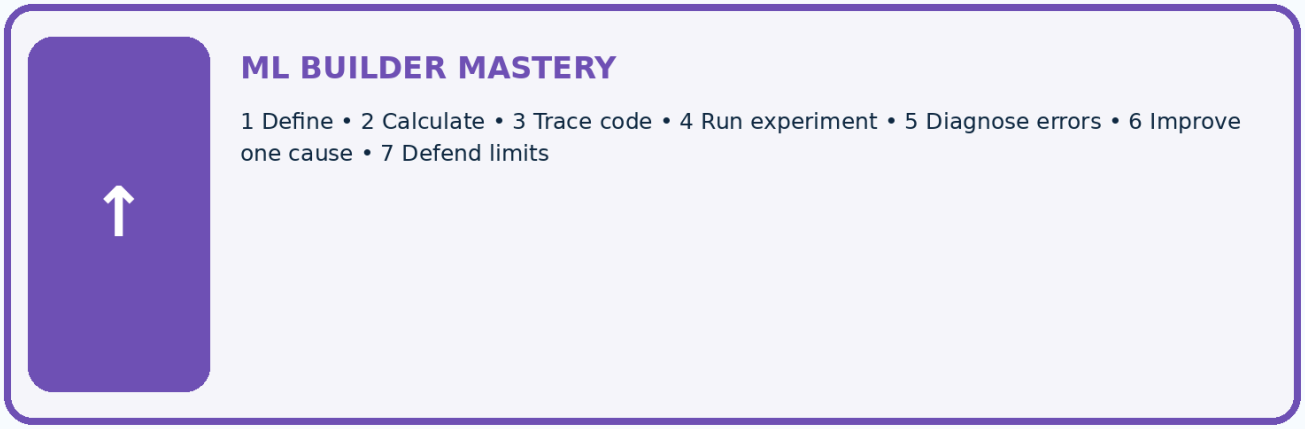
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Leakage. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **leakage**, **prediction time**, **post-outcome feature**, **split leakage**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A highly accurate feature should be kept even when it would not exist during real use.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

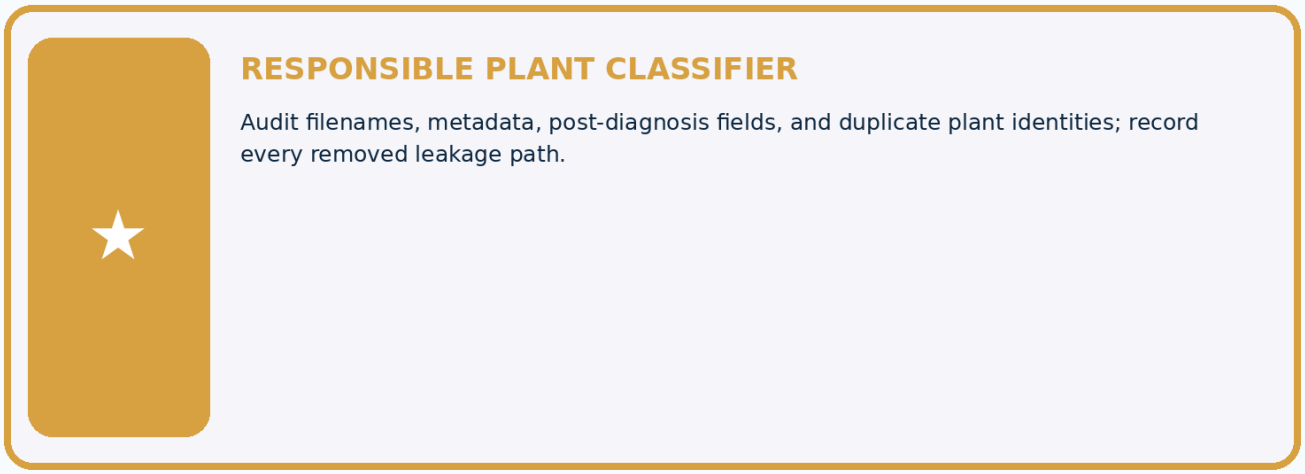
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Audit filenames, metadata, post-diagnosis fields, and duplicate plant identities; record every removed **leakage** path.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

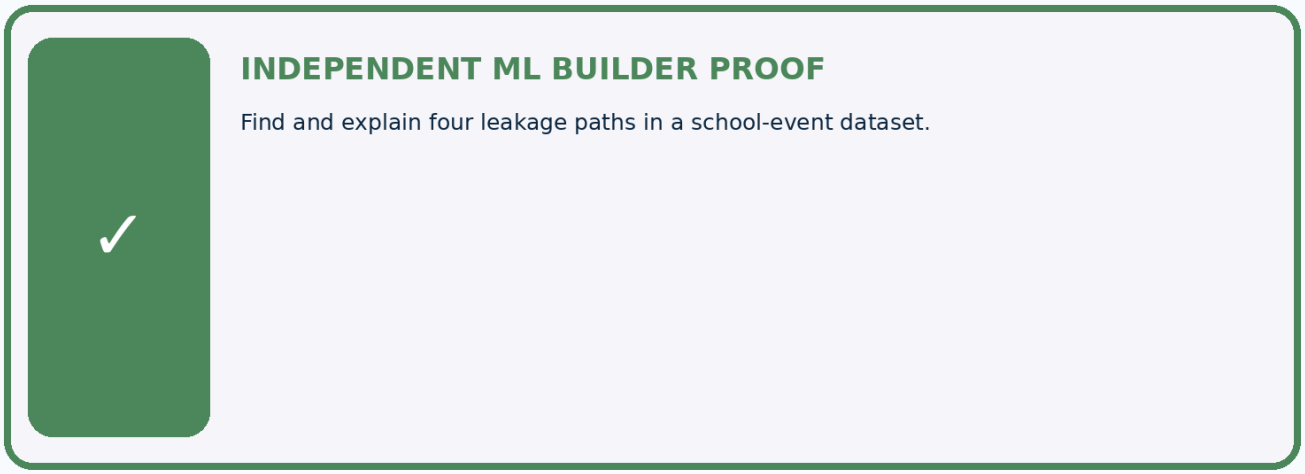
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Find and explain four **leakage** paths in a school-event dataset.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Dataset Cards.

Study a complete case

> Revisit the complete Leakage case for supported practice. Repeated examples do not become fresh assessments.

Week 8 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Ten photos per leaf are randomly split so one leaf appears in train and test.

The question

Explain and repair **leakage**.

Worked explanation

Related views make evaluation familiar. Split by leaf identity before **fitting** or learned pre**process**ing; keep all views together.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Keep related views in the same partition

Evidence or stage	Value or result
Leaf A, all views	10
Leaf B, all views	10
Leaf C, all views	10

Ten photos per leaf are randomly split so one leaf appears in train and test.

Explain and repair **leakage**.

Related views make evaluation familiar. Split by leaf identity before **fitting** or learned pre**process**ing; keep all views together.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Each of five plants has four photos. A random photo split puts different views of every plant in both **training** and **test set**s. Explain the problem and repair.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 8 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A feature records the expert's final species decision, made after the **prediction time**. Why is this **leakage**, and what should happen to it?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Audit filenames, metadata, post-diagnosis fields, and duplicate plant identities; record every removed **leakage** path.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Leakage, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Split by plant identity when testing generalisation to new plants.

By the end, I can	How I will show it
Choose a split unit that matches the intended prediction claim	Explain the deciding step, show a trace or test, and state one limit.
Keep related examples together and audit group overlap	Explain the deciding step, show a trace or test, and state one limit.
Record split roles counts and limits before model evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Leakage without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Choose a split unit that matches the intended **prediction** claim

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Split by plant identity when testing generalisation to new plants.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final test probes one unseen plant, not three independent plants.

1. Assign **group**s

2. State claim

3. **Group** related views

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: State claim The final test probes one unseen plant, not three independent plants. Read the picture in order.

The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move Fir's first view into **training** while retaining its other two in test. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The unseen-plant evaluation is compromised.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The unseen-plant evaluation is compromised. Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Repair a missing building block

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

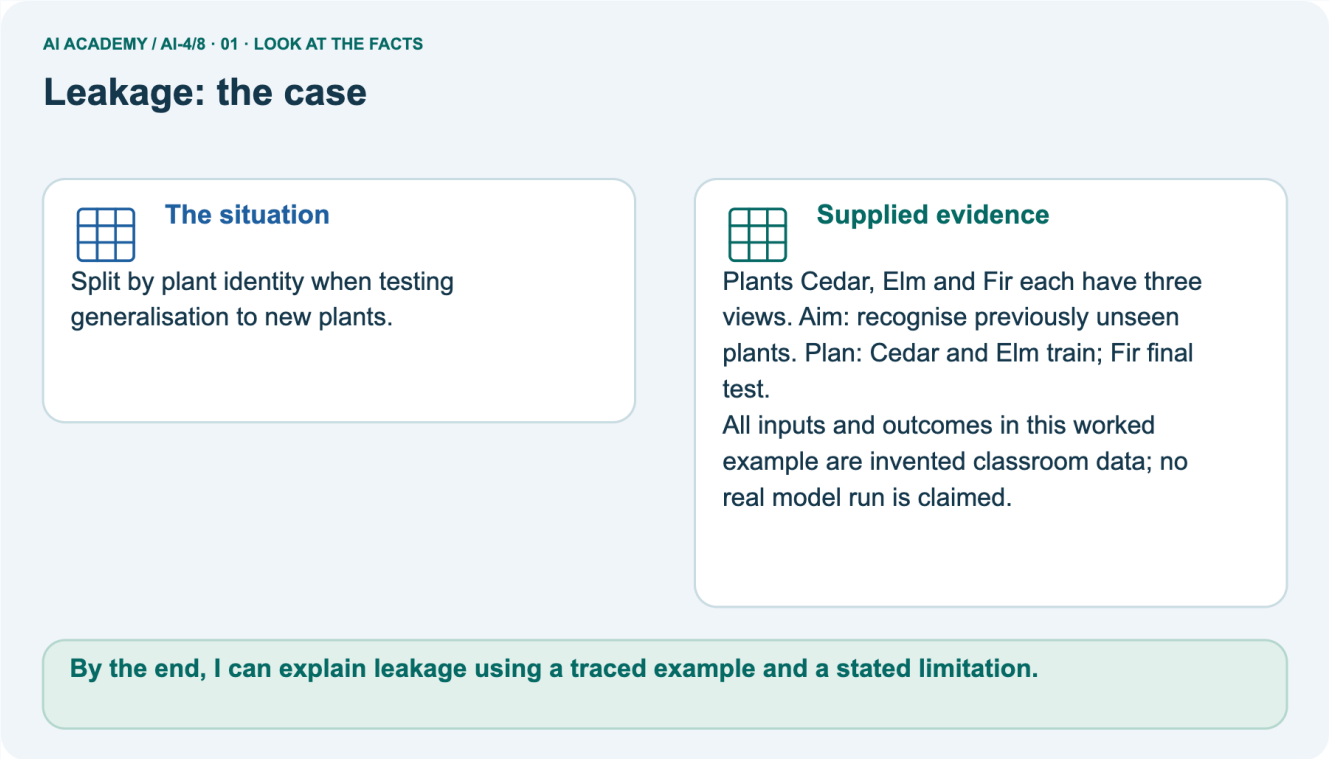
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Leakage graphic using its labels, deciding evidence and relevant limitation.

01. Leakage: the case



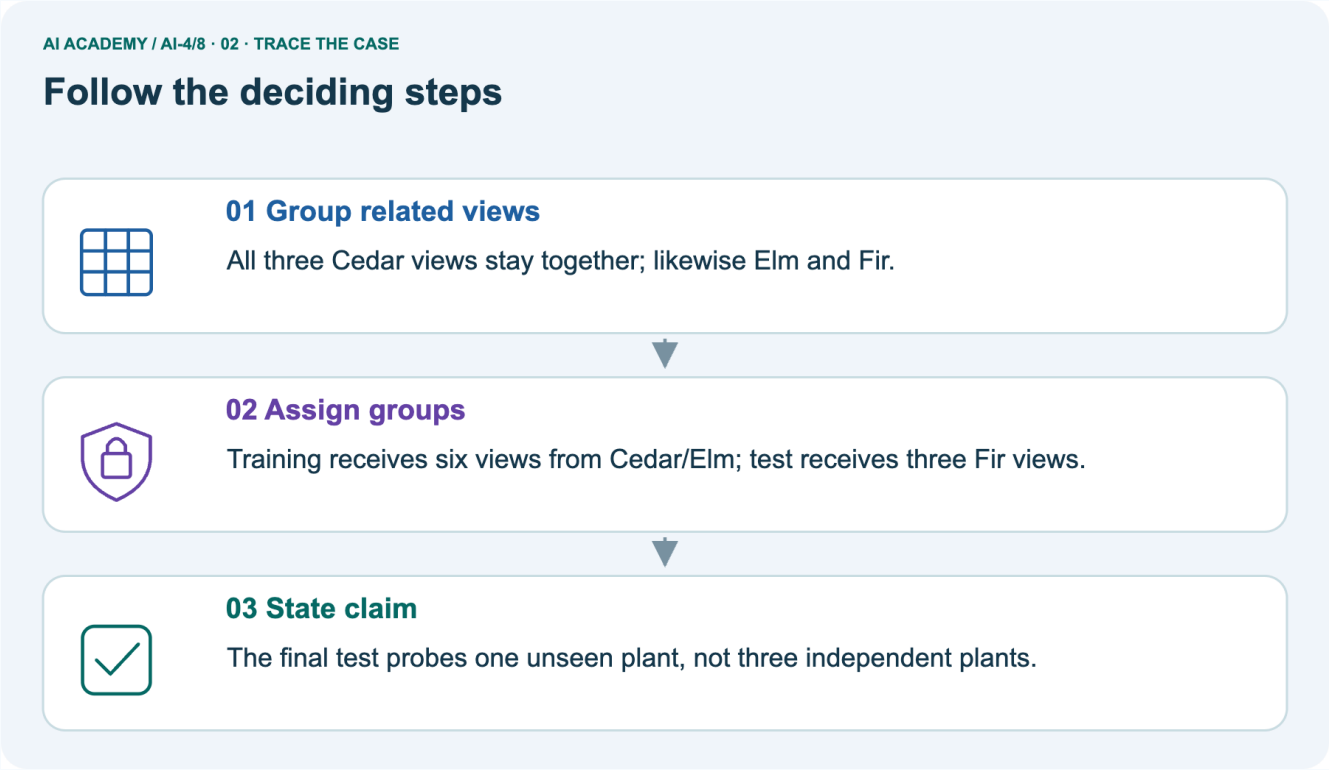
Follow the labelled picture with the matching explanation. Open the vector diagram

Split by plant identity when testing generalisation to new plants. Supplied facts: Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

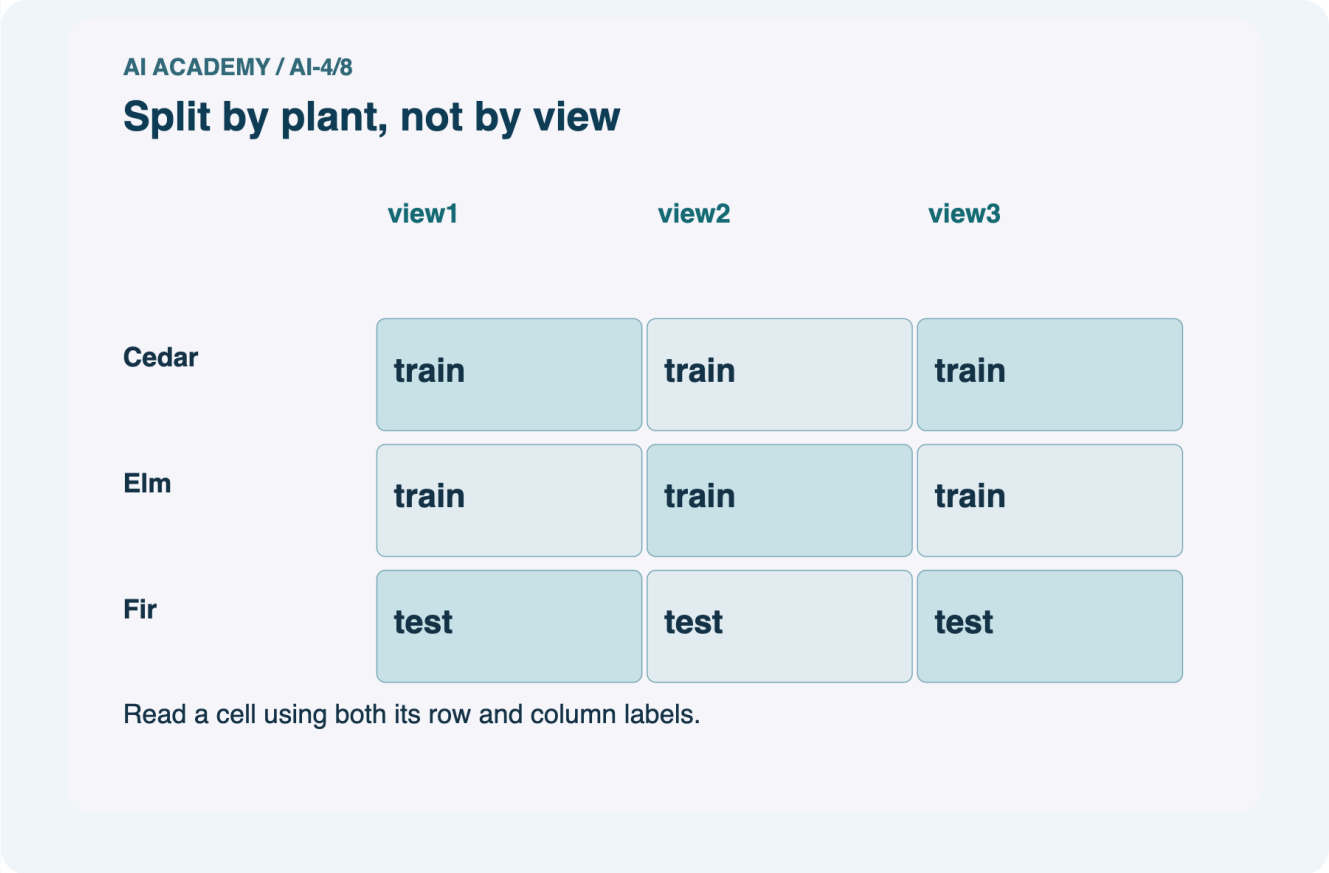


Follow the labelled picture with the matching explanation. Open the vector diagram

Group related views: All three Cedar views stay together; likewise Elm and Fir. → Assign **group**s: **Training** receives six views from Cedar/Elm; test receives three Fir views. → State claim: The final test probes one unseen plant, not three independent plants.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Split by plant, not by view

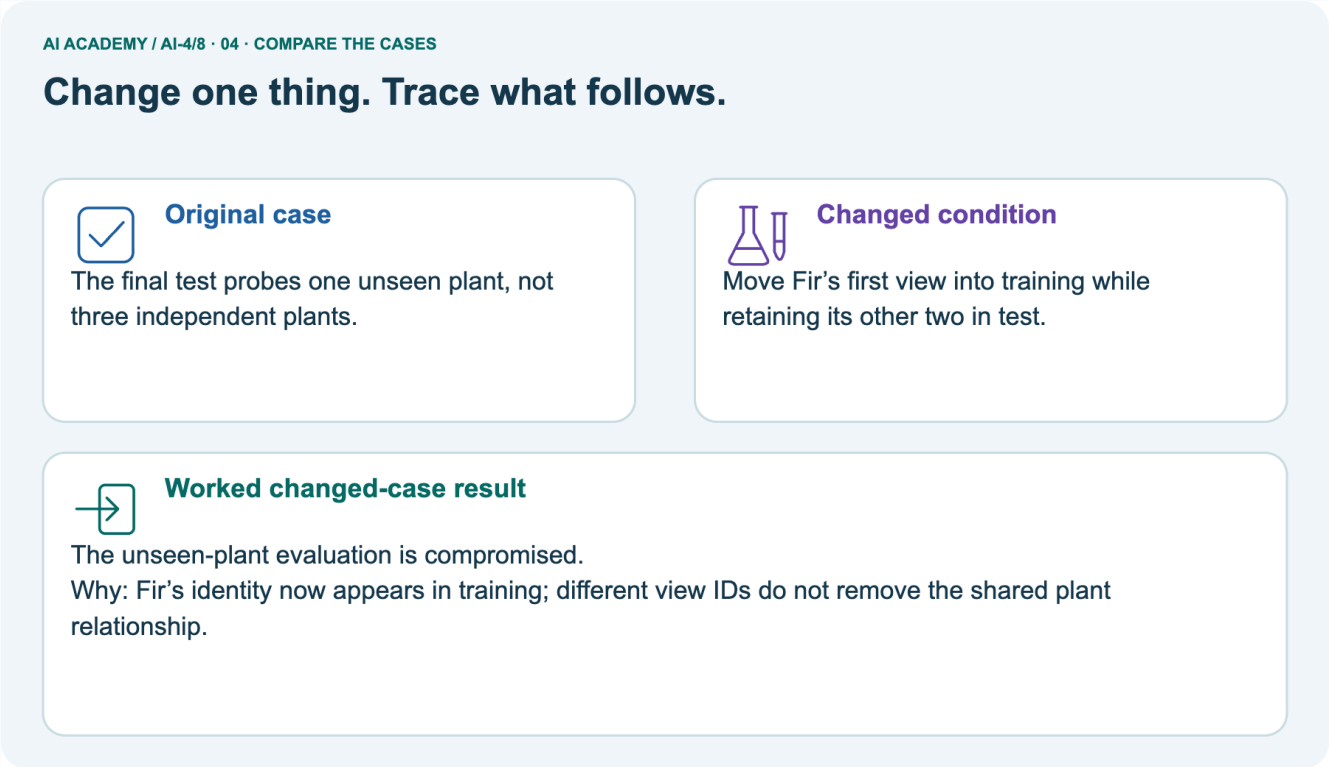


Follow the labelled picture with the matching explanation. Open the vector diagram

Split by plant, not by view. Cedar: view1 = train, view2 = train, view3 = train. Elm: view1 = train, view2 = train, view3 = train. Fir: view1 = test, view2 = test, view3 = test.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



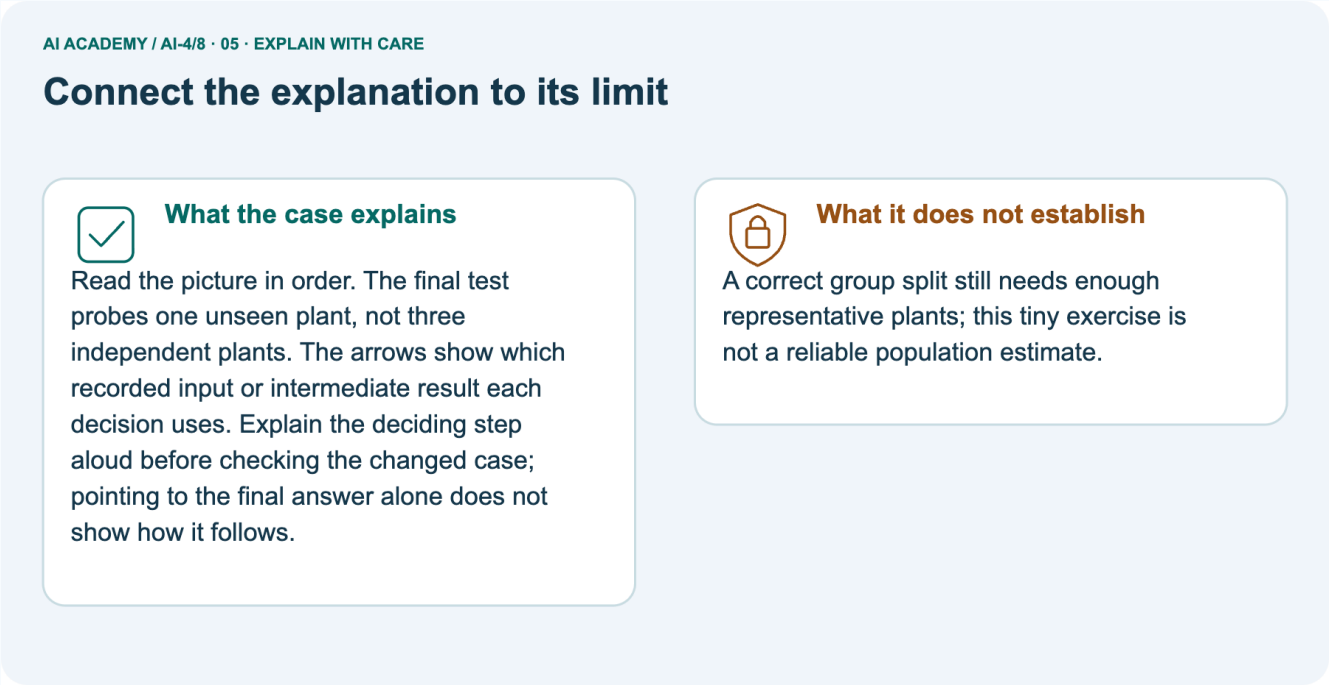
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final test probes one unseen plant, not three independent plants. Changed condition: Move Fir's first view into **training** while retaining its other two in test.

Expected result: The unseen-plant evaluation is compromised. Reason: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

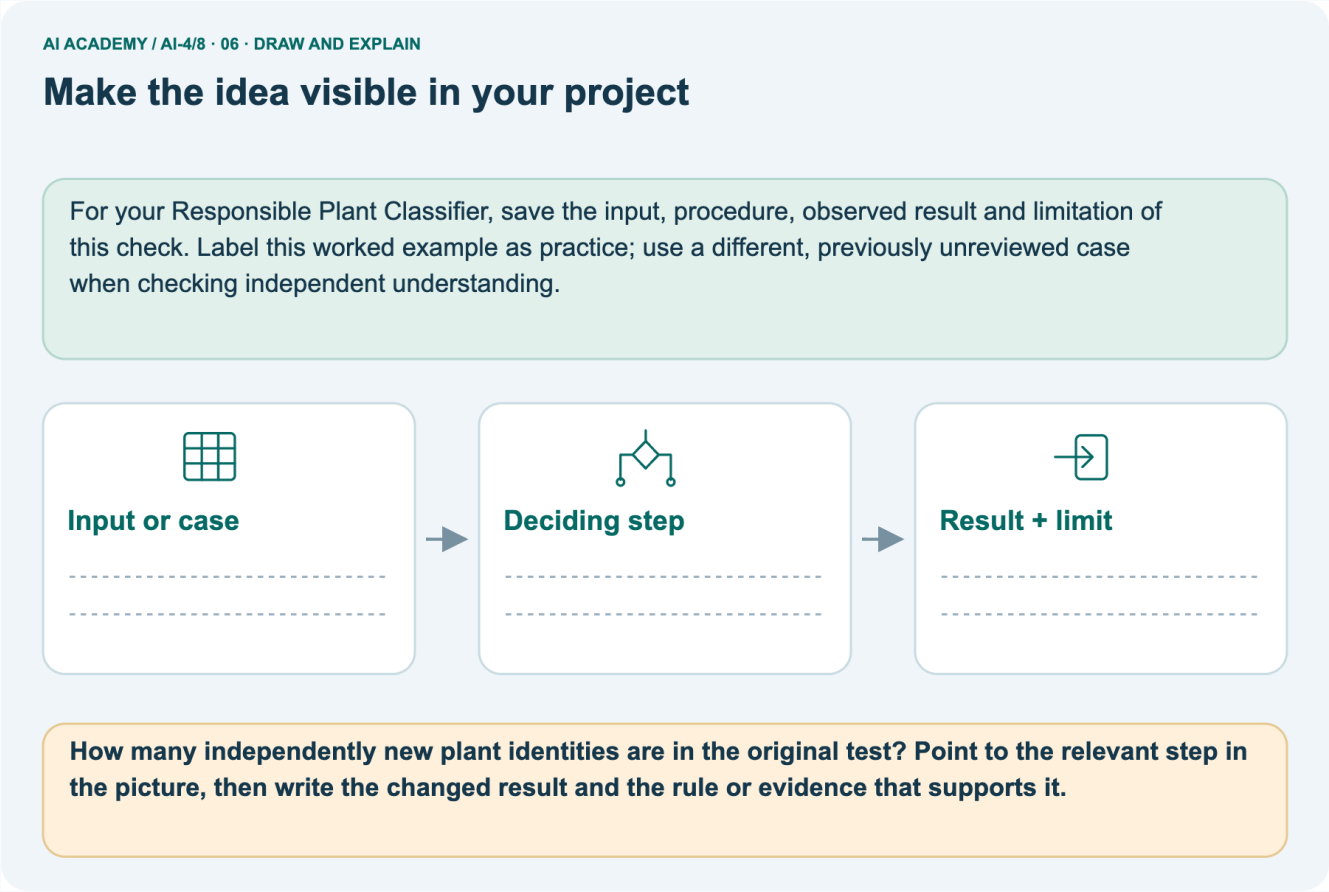
Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correct **group** split still needs enough representative plants; this tiny exercise is not a reliable population estimate.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.